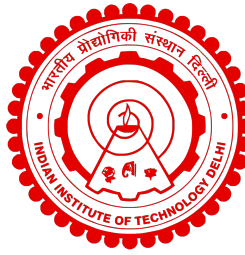
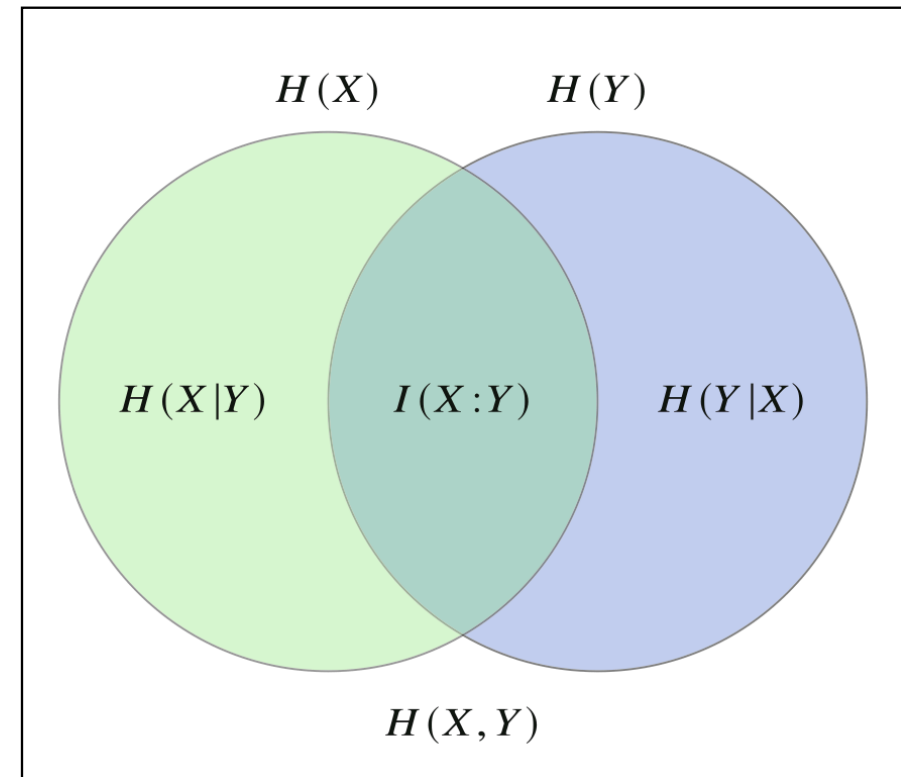


Relationship Between Entropies



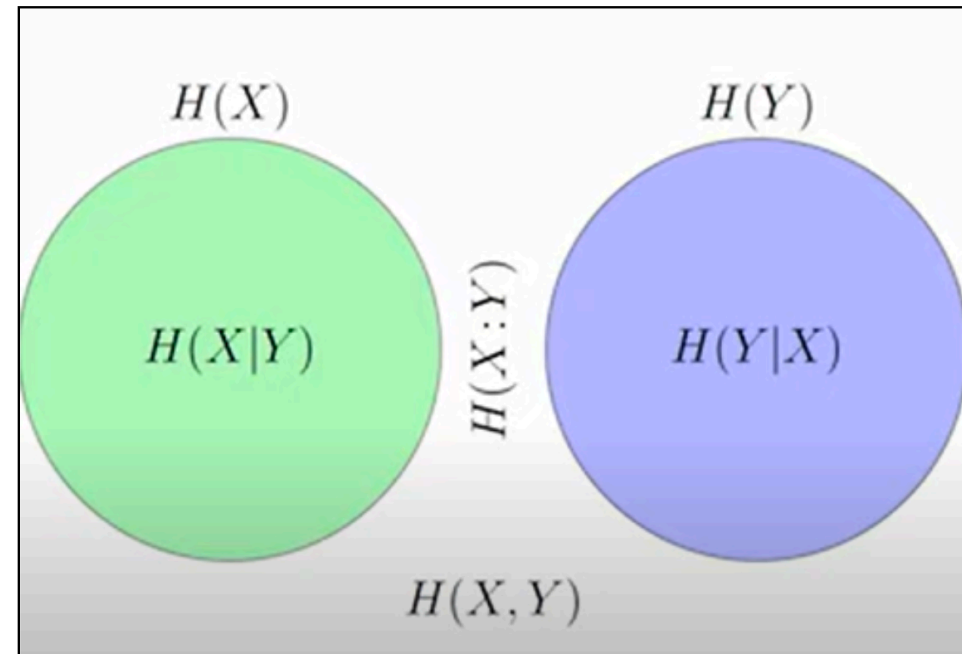
- $H(X|Y)$: Uncertainty in X after knowing Y (green without overlap).
- $H(Y|X)$: Uncertainty in Y after knowing X (blue without overlap).
- $I(X : Y)$: Mutual information, the shared information between X and Y .
- $H(X, Y)$: Total uncertainty in the joint system (X, Y) , equal to the union area.
- Relation: $H(X, Y) = H(X|Y) + I(X : Y) + H(Y|X)$



Mutual Information: Independence Case



- The disjoint green and blue circles represent the entropies of X and Y respectively.
- Since there is no overlap, the mutual information is: $I(X : Y) = 0$
- This means X and Y are independent.
- Consequently:
$$H(X|Y) = H(X), \quad H(Y|X) = H(Y)$$
- The joint entropy simplifies to: $H(X, Y) = H(X) + H(Y)$





Relative Entropy

- **Definition:** Suppose $\mathcal{X}$ is a finite set and $p(x)$ and $q(x)$ are two probability distributions over the set $\mathcal{X}$. The relative entropy of $p(x)$ to $q(x)$ is:

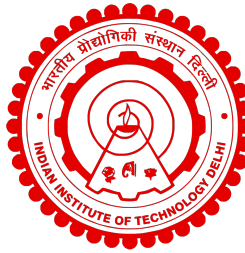
$$D(p(x) || q(x)) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right) = -H(X) - \sum_x p(x) \log(q(x))$$

- Where we define $-p(x) \log(0) = +\infty$ if $p(x) > 0$.
- **Theorem:** Given two probability distributions $p(x)$ and $q(x)$ over some alphabet $\mathcal{X}$, the relative entropy of $p(x)$ to $q(x)$ is non-negative:

$$D(p(x) || q(x)) \geq 0$$

- Equality holds if and only if $p(x) = q(x)$ for all $x \in \mathcal{X}$.

Example



p(x,y)	Y=0	Y=1
X=0	0.3	0.2
X=1	0.1	0.4

• Let X and Y be two binary random variables with the following joint probability distribution:

• **Joint Entropy:** $H(X, Y) = - \sum_{x,y} p_{X,Y}(x, y) \log(p_{X,Y}(x, y))$

$$H(X, Y) = - [0.3 \log_2 0.3 + 0.2 \log_2 0.2 + 0.1 \log_2 0.1 + 0.4 \log_2 0.4] \approx 1.846 \text{ bits}$$

• **Conditional Entropy:** $H(Y | X) = \sum_x P(x) H(Y | X = x)$

• Compute marginal: $P(X) : P(X = 0) = 0.3 + 0.2 = 0.5$,
 $P(X = 1) = 0.1 + 0.4 = 0.5$.

• Compute
 $H(Y | X = 0) = - [0.6 \log_2 0.6 + 0.4 \log_2 0.4] \approx 0.971$
 (since $P(Y | X = 0) = [0.6, 0.4]$)

• Similarly, $H(Y | X = 1) \approx 0.721$

• Result:
 $H(Y | X) = 0.5 \times 0.971 + 0.5 \times 0.721 \approx 0.846 \text{ bits}$.

• **Mutual Information:**
 $I(X : Y) = H(Y) - H(Y | X)$

• Compute:
 $H(Y) : P(Y = 0) = 0.3 + 0.1 = 0.4, P(Y = 1) = 0.6$,
 $H(Y) = - [0.4 \log_2 0.4 + 0.6 \log_2 0.6] \approx 0.971 \text{ bits}$

• Result: $I(X; Y) = 0.971 - 0.846 = 0.125 \text{ bits}$

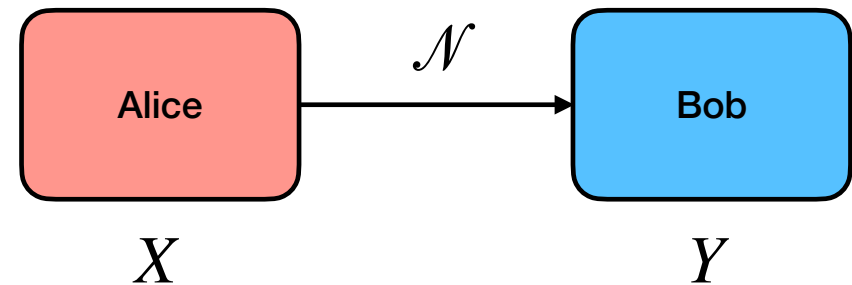


Shannon Channel Capacity

- Consider a noisy classical communication channel $\mathcal{N}$. Let:
 - X : input random variable (Alice's side)
 - Y : output random variable (Bob's side)
- The **capacity** of the channel is:

$$C(\mathcal{N}) = \max_{p_X(x)} I(X : Y)$$

- Optimized over all input distributions $p_X(x)$.
- Represents max information per channel use.



Shannon's Second Question: Channel Capacity



- **Q2:** *How much information can be reliably transmitted through a noisy communication channel?*
- **A2:** *According to **Shannon's Channel Coding Theorem**, the maximum reliable transmission rate is called the **Channel Capacity**:*

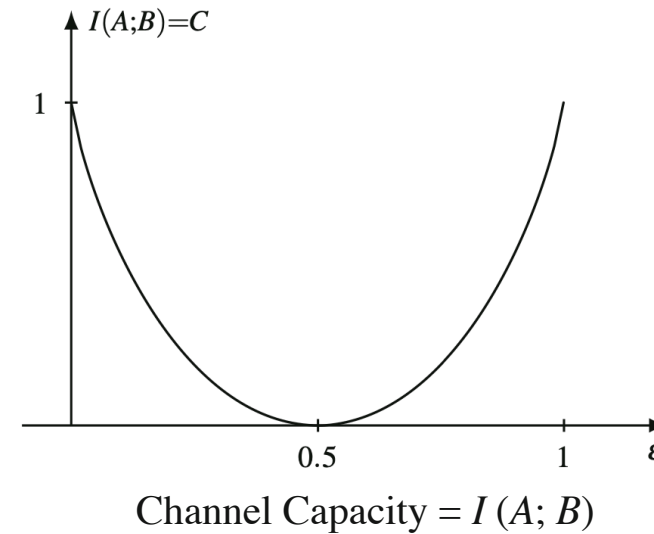
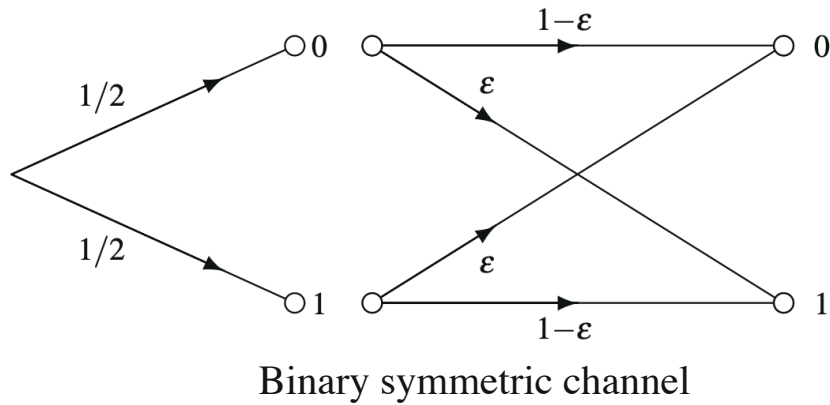
$$C(\mathcal{N}) = \max_{p_X(x)} I(X : Y)$$

- **Explanation:**
 - Mutual Information $I(X : Y) = H(X) - H(X|Y)$ measures how much knowing Y reveals about X .
 - Capacity $C(\mathcal{N})$ represents the highest number of bits per channel use that can be transmitted with an arbitrarily low probability of error.
 - The maximization is over all input distributions $p_X(x)$, because different input strategies yield different efficiencies in using the channel.



Channel Capacity Example

- **Binary symmetric channel:** Cross transition probabilities: $p_c(0|1) = p_c(1|0) = \varepsilon$
- For equiprobable symbols $p_A(0) = p_A(1) = 1/2$, source entropy is $H(A) = 1$ bit/symbol
- Equivocation: $H(A|B) = -(1-\varepsilon)\log(1-\varepsilon) - \varepsilon\log\varepsilon$
- Mutual information: $I(A;B) = H(A) - H(A|B) = 1 + (1-\varepsilon)\log(1-\varepsilon) + \varepsilon\log\varepsilon$





Shannon's Noisy Channel Coding

- Channel coding adds suitable redundancy in order to retrieve the original message with a small error probability (reliable communication)
- Coding in the Presence of Noise:

$A \in \mathcal{A} = \{1, \dots, K\}$ symbol (or message) produced by the source,
 $X^L = (X_1 X_2 \dots X_L) \in \mathcal{C}$ codeword sent to the channel,
 $Y^L = (Y_1 Y_2 \dots Y_L) \in \mathcal{Y}^L$ received codeword,
 $\hat{A} \in \mathcal{A}$ decoded symbol (or message).

- Channel Encoder: $x^L = g(a)$, $a \in \mathcal{A}$, $x^L \in \mathcal{C}$
- L -extension of channel $(\mathcal{X}^L, \mathcal{Y}^L, p_{cL})$, where:
$$p_{cL}(y_1 y_2 \dots y_L | x_1 x_2 \dots x_L) = \prod_{n=1}^L p_c(y_n | x_n)$$
- Channel Decoder: $\mathcal{D} : \mathcal{Y}^L \rightarrow \mathcal{A}$, where
 $\hat{a} = \mathcal{D}(y^L)$, $y^L \in \mathcal{Y}^L$, $\hat{a} \in \mathcal{A}$

